

HOMWORK DAY 9 - Chain Rule §2.5

1. §2.5: 7. $f(x) = (2x^3 - 5x^2 + 4)^5$ $f'(x) = 10x(3x-5)(2x^3 - 5x^2 + 4)$
 $f'(x) = 5(2x^3 - 5x^2 + 4)^4 \cdot \frac{d}{dx}(2x^3 - 5x^2 + 4)$
 $f'(x) = 5(2x^3 - 5x^2 + 4)^4 \cdot (6x^2 - 10x)$
 $= 30x^2 - 50x(2x^3 - 5x^2 + 4)$

2. §2.5: 9. $f(x) = \sqrt{5x+1}$ $f'(x) = \frac{1}{2}(5x+1)^{-1/2} \cdot \frac{d}{dx}(5x+1)$ $f'(x) = \frac{5}{2\sqrt{5x+1}}$
 $= (5x+1)^{1/2}$
 $= \frac{1}{2}(5x+1)^{-1/2} \cdot 5$
 $= \frac{5}{2}(5x+1)^{-1/2}$

3. §2.5: 10. $f(x) = \frac{1}{\sqrt[3]{x^2-1}}$ $f'(x) = -\frac{1}{3}(x^2-1)^{-4/3} \cdot \frac{d}{dx}(x^2-1)$
 $= \frac{1}{(x^2-1)^{1/3}}$
 $= (x^2-1)^{-1/3}$ $f'(x) = -\frac{2x}{3\sqrt[3]{(x^2-1)^4}}$
 $f'(x) = -\frac{2x}{3\sqrt[3]{(x^2-1)^4}}$

4. §2.5: 14. $g(x) = (2 - \sin x)^{3/2}$ $g'(x) = \frac{3}{2}(2 - \sin x)^{1/2} \cdot \frac{d}{dx}(2 - \sin x)$ $g'(x) = \frac{3 \cos x (2 - \sin x)^{1/2}}{2}$
 $= \frac{3}{2}(2 - \sin x)^{1/2} \cdot -\cos x$

5. §2.5: 15. $f(\theta) = \cos(\theta^2)$
 $f'(\theta) = -\sin(\theta^2) \cdot \frac{d}{d\theta} \theta^2$
 $= -\sin(\theta^2) \cdot 2\theta$
 $f'(\theta) = -2\theta \sin(\theta^2)$

6. §2.5: 16. $g(\theta) = \cos^2 \theta$ $= 2(\cos \theta) \cdot -\sin \theta$
 $= (\cos \theta)^2$ $g'(\theta) = -2 \sin \theta (\cos \theta)$
 $g'(\theta) = 2(\cos \theta) \cdot \frac{1}{2} \cos \theta$ $g'(\theta) = -2 \sin 2\theta$

7. §2.5: 18. $f(t) = t \sin(\pi t)$

8. §2.5: 24. $y = (x + \frac{1}{x})^5$ $y' = 5(x + x^{-1})^4 \cdot \frac{d}{dx}(x + x^{-1})$
 $y = (x + x^{-1})^5$ $= 5(x + x^{-1})^4 \cdot (-x^{-2} + 1)$
 $= 5(x + \frac{1}{x})^4 (1 - \frac{1}{x^2})$

9. Let $f(t) = (3t - 1)^4(2t + 1)^{-3}$.

(a) Find $f'(t)$. Simplify your answer

Product rule $f(x) \cdot g(x) = f'(x)g(x) + g'(x)f(x)$

$$\begin{aligned} f'(t) &= [4(3t-1)^3 \cdot \frac{d}{dt}(3t-1) \cdot (2t+1)^{-3}] + [-3(2t+1)^{-4} \cdot \frac{d}{dt}(2t+1) \cdot (3t-1)^4] \\ &= [4(3t-1)^3 \cdot 3 \cdot (2t+1)^{-3}] + [-3(2t+1)^{-4} \cdot 2 \cdot (3t-1)^4] \\ &= [12(3t-1)^3 \cdot (2t+1)^{-3}] + [-6(2t+1)^{-4} \cdot (3t-1)^4] \end{aligned}$$

$$f'(t) = \frac{12(3t-1)^3}{(2t+1)^3} - \frac{6 \cdot (3t-1)^4}{(2t+1)^4}$$

(b) Find the points (a, b) on the curve $y = f(t)$ at which the tangent line is horizontal.

$$f'(t) = 0 \quad \left(\frac{12(3t-1)^3}{(2t+1)^3} - \frac{6(3t-1)^4}{(2t+1)^4} = 0 \right)$$

$$12(3t-1)^3(2t+1) - 6(3t-1)^4 = 0$$

$$6(t+3)(3t-1)^3 = 0$$

$$t+3=0, 3t-1=0$$

$$t=-3, t=\frac{1}{3}$$

$$\begin{aligned} f(-3) &= (3(-3)-1)^4(2(-3)+1)^{-3} \\ &= (-10)^4 \cdot 5^{-3} \\ &= 10000 \cdot \frac{1}{125} \\ &= \frac{10000}{125} = \frac{200}{25} = \frac{40}{5} = 8 \end{aligned}$$

$$\begin{aligned} f\left(\frac{1}{3}\right) &= \left(3\left(\frac{1}{3}\right)-1\right)^4 \cdot \left(2\left(\frac{1}{3}\right)+1\right)^{-3} \\ &= \left(\frac{2}{3}+1\right)^{-3} \\ &= \frac{1}{\left(\frac{5}{3}\right)^3} = \frac{3}{5} \end{aligned}$$

POINTS

$$(-3, 8)$$

$$\left(\frac{1}{3}, \frac{3}{5}\right)$$

10. Let $f(x) = \frac{1}{4+x^2}$

(a) Find $f'(x)$ (simplify)

$$f(x) = (4+x^2)^{-1}$$

$$f'(x) = -1(4+x^2)^{-2} \cdot \frac{d}{dx}(4+x^2)$$

$$= -1(4+x^2)^{-2} \cdot 2x$$

$$= -2x(4+x^2)^{-2}$$

$$f'(x) = -\frac{2x}{(4+x^2)^2}$$

(b) Find $f''(x)$ (simplify)

constant rule $f'(x) = -\frac{2x}{(4+x^2)^2} = -2 \cdot \frac{x}{(4+x^2)^2}$

product rule $f'(x) = -2 \cdot \frac{d}{dx} \left(\frac{x}{(4+x^2)^2} \right)$

$$f''(x) = -2 \cdot \frac{[(4+x^2)^2 \cdot 1] - [x \cdot 2(4+x^2) \cdot 2x]}{[(4+x^2)^2]^2}$$

$$= -2 \cdot \frac{(4+x^2)^2 - 4x^2 \cdot 2(4+x^2)}{(4+x^2)^4}$$

$$= -2 \cdot \frac{(4+x^2)^2 - 8x^2(4+x^2)}{(4+x^2)^4}$$

$$f''(x) = -\frac{2(-3x^2+4)}{(4+x^2)^3}$$

(c) Find the points (a,b) on the curve $y = f(x)$ at which $f''(x) = 0$.

Solve, $f''(x) = 0$

$$\frac{2(-3x^2+4)}{(4+x^2)^3} = 0$$

D: All #s, therefore if $\frac{f'(x)}{g(x)}$, solve for $f(x)=0$

$$\frac{2(-3x^2+4)}{2} = 0$$

$$-3x^2+4=0$$

$$-3x^2 = -4$$

$$\sqrt{x^2} = \sqrt{\frac{4}{3}}$$

$$x = \pm \sqrt{\frac{2}{3}}$$

$$x = -\frac{2}{\sqrt{3}}, \frac{2}{\sqrt{3}}$$

$$f\left(\frac{2}{\sqrt{3}}\right) = \frac{1}{4+\left(\frac{2}{\sqrt{3}}\right)^2}$$

$$= \frac{1}{4+\frac{4}{3}}$$

$$= \frac{1}{\frac{16}{3}} = \frac{3}{16}$$

$$f\left(\frac{2}{\sqrt{3}}\right) = \frac{3}{16}$$

POINTS

$$\left(-\frac{2}{\sqrt{3}}, \frac{3}{16}\right) \text{ and } \left(\frac{2}{\sqrt{3}}, \frac{3}{16}\right)$$

11. §2.5: 66

x	f	f'	g	g'
1	3	4	2	6
2	1	5	8	7
3	7	7	2	9

a) $F(x) = f(f(x))$

$$F'(x) = f'(f(x)) \cdot f'(x)$$

$$F'(2) = f'(f(2)) \cdot f'(2)$$

$$= f'(1) \cdot 5$$

$$= 4 \cdot 5 = 20$$

b) $G(x) = g(g(x))$

$$G'(x) = g'(g(x)) \cdot g'(x)$$

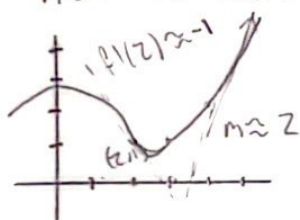
$$G'(3) = g'(g(3)) \cdot g'(3)$$

$$= g'(2) \cdot 9$$

$$= 7 \cdot 9 = 63$$

12. §2.5: 68

$h(x) = f(f(x))$, $g(x) = f(x^2)$



a) $h'(z) = f'(f(z)) \cdot f'(z)$

$$= f'(2) \cdot f'(2)$$

$$= -1 \cdot -1 = 1$$

b) $g(z) = f(z^2)$

$$g'(z) = f'(4)$$

$$= 2$$

13. §2.5: 72 $f(x) = xg(x^2)$, find f'' in terms of g, g' and g''

$$f'(x) = g(x^2) + 2x^2 \cdot g'(x^2)$$

$$f''(x) = 2xg'(x^2) + 4x^2 \cdot g'(x^2) + 2xg''(x^2) \cdot 2x^2$$

$$= 6xg'(x^2) + 4x^3g''(x^2)$$

14. §2.5: 78

$s = A \cos(\omega t + \delta)$; A, ω, δ are all constants He for $y' = 0$

a) $\frac{ds}{dt} = -A \sin(\omega t + \delta) \cdot \omega$

$\frac{ds}{dt} = -A \omega \sin(\omega t + \delta)$

b) $0 = \frac{ds}{dt}$, solve

$-A \omega \sin(\omega t + \delta) = 0$ velocity = 0 when $\omega t + \delta = 0$ or π

$\omega t + \delta = 0$, $\omega t + \delta = \pi$

$t = -\frac{\delta}{\omega}$, $t = \frac{\pi - \delta}{\omega}$

15. §2.5: 81

$a(t) = v(t) \frac{dv}{ds}$, $\frac{a(t)}{v(t)} = \frac{dv}{ds}$

- $a(t)$ measures the change in velocity.
- $v(t)$ measures the change in displacement.

So therefore $\frac{dv}{dt}$ measures the change in velocity w/ respect to time
whereas, $\frac{dv}{ds}$ measure change in velocity w/ respect to displacement.